

THE COLA-URINE DETECTIVE AGENCY — NEPHRITIC SYNDROME.

SUSPECT LINEUP

- SPER bomb
- PSGN
- IgA
- HSP / IgA VASCULITIS
- LUPUS
- ANTI-GBM GLOMERULOPATHY
- ANCA TROPIC GN
- ALPORT
- GRAY AREA — CAN BE NEPHROTIC OR NEPHRITIC

PENTAD.

- HEMATURIA + RBC CASTS + DYSMORPHIC RBCs
- SUBNEPHROTIC PROTEINURIA
- HYPERTENSION
- EDEMA — less than nephrotic
- AKI / OLIGURIA

COMPLEMENT CLUE BOARD

↓C3 + NORMAL C4 (ALTERNATIVE PATHWAY)	↓C3 + ↓C4 (CLASSICAL PATHWAY)	NORMAL C3 + C4
PSGN	LUPUS	IgA
MPGN type II / CSG	CRYOGLOBULINEMIA	ANTI-GBM
atypical HUS	MPGN type I	ANCA
	POST-INFECTION ENDOCARDITIS	THIN BM
		ALPORT

WORKUP TOOLBOX

- ANCA
- ANTI-GBM
- STRIP-GRAPE
- STRIP-LEMON
- STRIP-PURPLE
- STRIP-BLUE
- STRIP-RED
- STRIP-YELLOW
- STRIP-ORANGE
- STRIP-GREEN
- STRIP-PINK
- STRIP-VIOLET
- STRIP-BROWN
- STRIP-SILVER
- STRIP-GOLD
- STRIP-COPPER
- STRIP-CRICKET
- STRIP-DIAMOND
- STRIP-EMERALD
- STRIP-FELDSPAR
- STRIP-FELDSPATHOIDS
- STRIP-GARNET
- STRIP-JEWELLERY
- STRIP-KAPORITE
- STRIP-LAZULITE
- STRIP-MAGNETITE
- STRIP-MALACHITE
- STRIP-MOLYBDENITE
- STRIP-NATROLITE
- STRIP-OBSIDIAN
- STRIP-OPAL
- STRIP-QUARTZ
- STRIP-RUBY
- STRIP-SAPPHIRE
- STRIP-TOPAZ
- STRIP-TURQUOISE
- STRIP-VANADIUM
- STRIP-WHITE
- STRIP-YELLOW
- STRIP-ZINC
- STRIP-ZIRCON

RPGN — RAPID GFR DECLINE / EMERGENCY

- TYPE I — LINEAR IgG Anti GBM
- TYPE II — GRANULAR Immune Complex Anti IgA, PSGN, MPGN, etc.
- TYPE III — PAUCI-INSANE ANCA

URGENT TX — PLEX STEROIDS CYCLO PHOSPHAMIDE

1. Hematuria / RBC casts

WHAT YOU SEE

Cola-colored urine cup + RBC cast in suspect lineup

WHAT IT ENCODES

Hematuria with DYSMORPHIC red cells (acanthocytes) and especially RED-CELL CASTS is the defining hallmark of nephritic syndrome — it proves the bleeding is GLOMERULAR (cells squeeze through an inflamed, leaky GBM and mold into casts within tubules). Urine is classically 'cola/tea-colored' or smoky brown. RBC casts are essentially pathognomonic for glomerulonephritis.

BOARD TRAP

WRONG: 'Bland urinary sediment with oval fat bodies / Maltese crosses indicates nephritic syndrome.' Discriminator — oval fat bodies, fatty casts, and a BLAND sediment are NEPHROTIC findings. Nephritic urine shows an ACTIVE sediment: dysmorphic RBCs and RBC casts. Casts = nephritic; lipiduria = nephrotic.

2. Sub-nephrotic proteinuria

WHAT YOU SEE

A suspect placard reading SUB-NEPHROTIC PROTEINURIA in the mug-shot lineup, holding a small (not overflowing) protein cup — the modest spill marks <3.5 g/day, distinguishing nephritic from heavy nephrotic loss.

WHAT IT ENCODES

Proteinuria IS present in nephritic syndrome but is sub-nephrotic — typically <3.5 g/day (often 1–2 g/day) — because the injury is inflammatory/cellular rather than a pure podocyte filtration-barrier defect. Heavy 'nephrotic-range' protein loss with hypoalbuminemia and edema points instead toward nephrotic syndrome.

BOARD TRAP

WRONG: 'Proteinuria >3.5 g/day with hypoalbuminemia is expected in nephritic syndrome.' Discriminator — >3.5 g/day defines NEPHROTIC range. Pure nephritic disease has sub-nephrotic proteinuria (<3.5 g); when proteinuria is heavy AND there is an active sediment, think OVERLAP disease (MPGN, diffuse proliferative lupus, IgA).

3. Hypertension

WHAT YOU SEE

Pentad photo — angry red faces (HTN)

WHAT IT ENCODES

A falling GFR causes salt and water RETENTION → intravascular volume expansion → HYPERTENSION. This is a core nephritic feature (part of the classic pentad) and helps distinguish it from the more variable BP in nephrotic disease. BP control with salt restriction, loop diuretics, and RAAS-sparing antihypertensives is part of management.

BOARD TRAP

WRONG: 'Nephritic syndrome causes hypotension from fluid loss.' Discriminator — nephritic disease RETAINS fluid (low GFR → Na/water retention), producing HYPERTENSION and volume overload, not hypotension. Hypotension would suggest volume depletion or sepsis, not glomerulonephritis.

4. Edema

WHAT YOU SEE

Pentad — puffy edema face

WHAT IT ENCODES

Edema in nephritic syndrome is typically MILD and periorbital/facial, driven by primary renal Na/water retention from low GFR (a 'nephritic/overflow' mechanism), not by hypoalbuminemia. It is less severe than nephrotic edema.

BOARD TRAP

WRONG: 'Massive generalized anasarca from severe hypoalbuminemia is the typical edema of nephritic syndrome.' Discriminator — massive anasarca from low oncotic pressure (hypoalbuminemia) is the NEPHROTIC mechanism. Nephritic edema is milder and results from salt/water retention secondary to reduced GFR.

5. AKI / Oliguria / Azotemia

WHAT YOU SEE

Pentad — kidney with oliguria label

WHAT IT ENCODES

Inflammatory glomerular injury drops the GFR, producing OLIGURIA and AZOTEMIA (rising BUN and creatinine) — i.e., an acute nephritic AKI. A rapid decline over days–weeks signals RPGN/crescentic disease and is an emergency.

BOARD TRAP

WRONG: 'Polyuria with a rising GFR is characteristic of acute nephritic syndrome.' Discriminator — nephritic injury LOWERS GFR, causing oliguria and azotemia, not polyuria. Polyuria suggests tubular concentrating defects (e.g., recovering ATN, diabetes insipidus), not glomerulonephritis.

6. Low complement causes

WHAT YOU SEE

The 'Complement Clue Board' column with downward C3 arrows listing PSGN, lupus, MPGN, cryoglobulinemia, endocarditis — the dropped complement (consumed by immune complexes) is the visual link tying these causes together.

WHAT IT ENCODES

LOW-complement nephritic causes (immune-complex/consumptive) — memory aid: post-streptococcal GN, lupus nephritis, MPGN (types I/II/III), cryoglobulinemia (often HCV-related), and subacute bacterial ENDOCARDITIS-associated GN. The low complement reflects activation/consumption of the cascade by immune complexes.

BOARD TRAP

WRONG: 'Anti-GBM disease and ANCA-associated vasculitis cause low complement.' Discriminator — pauci-immune (ANCA) and anti-GBM disease have NORMAL complement. Low complement points to immune-complex disease (PSGN, lupus, MPGN, cryoglobulinemia, endocarditis). Checking C3/C4 is the fastest way to split the nephritic differential.

7. Normal complement causes

WHAT YOU SEE

The clue-board column marked NORMAL C3/C4 listing ANCA vasculitis, anti-GBM, IgA/HSP — unchanged complement levels (no consumption) visually separate this pauci-immune/anti-GBM group from the low-complement causes.

WHAT IT ENCODES

NORMAL-complement nephritic causes (pauci-immune / anti-GBM / mesangial-IgA group): ANCA-associated vasculitis (GPA, MPA, EGPA — pauci-immune), anti-GBM / Goodpasture disease, IgA nephropathy, and Henoch–Schönlein purpura (IgA vasculitis). These are NOT driven by classical/alternate complement consumption, so C3/C4 stay normal.

BOARD TRAP

WRONG: 'Post-streptococcal GN belongs in the normal-complement column.' Discriminator — PSGN consumes complement and has LOW C3 (typically normalizing by ~8 weeks). Placing PSGN with the normal-complement diseases is a classic error; PSGN, lupus, MPGN, cryo, and endocarditis are the LOW-complement group.

8. Nephritic-nephrotic overlap

WHAT YOU SEE

Gray Area placard — overlap can be both

WHAT IT ENCODES

Nephritic and nephrotic are ends of a spectrum, and several diseases STRADDLE both — classically MPGN, diffuse proliferative lupus nephritis (class IV), and sometimes IgA nephropathy — presenting with active sediment/RBC casts AND nephrotic-range proteinuria simultaneously. Recognizing overlap prevents misclassification.

BOARD TRAP

WRONG: 'Nephritic and nephrotic are mutually exclusive categories; a patient cannot have both.' Discriminator — overlap is real and tested: MPGN and diffuse proliferative lupus commonly produce a mixed nephritic–nephrotic picture (RBC casts plus >3.5 g proteinuria). Treating the two as rigid boxes leads to missed diagnoses.

9. RPGN / crescents

WHAT YOU SEE

Red emergency ER wing — RPGN rapid GFR decline

WHAT IT ENCODES

Rapidly progressive glomerulonephritis (RPGN) = nephritic syndrome with GFR loss over DAYS-TO-WEEKS and CRESCENTS (proliferating parietal epithelial cells, macrophages, fibrin) on biopsy. It is a nephrologic EMERGENCY. Three immunologic patterns: type I anti-GBM (linear IF), type II immune-complex (granular IF — PSGN/lupus/IgA), type III pauci-immune (ANCA, scant staining). Needs urgent biopsy and immunosuppression (steroids + cyclophosphamide/rituximab ± plasmapheresis for anti-GBM).

BOARD TRAP

WRONG: 'RPGN follows a slow, indolent course and can be worked up electively over months.' Discriminator — RPGN is defined by RAPID GFR decline (days–weeks) and demands EMERGENT biopsy and immunosuppression; delay leads to irreversible crescentic scarring and ESKD. An indolent course argues against RPGN.

10. Urinalysis / microscopy

WHAT YOU SEE

Microscope + cola urine cup on detective desk

WHAT IT ENCODES

The FIRST step in any glomerular workup is URINALYSIS with sediment microscopy — looking for dysmorphic RBCs, RBC casts, and quantifying proteinuria (spot protein:creatinine ratio). This cheap, fast test localizes the lesion (nephritic vs nephrotic) and directs subsequent serologies and biopsy.

BOARD TRAP

WRONG: 'Proceed directly to renal biopsy before checking a urinalysis.' Discriminator — urinalysis/microscopy is step ONE; it confirms glomerular hematuria (RBC casts) and guides the rest of the workup. Jumping to invasive biopsy before a UA and serologies skips the highest-yield, lowest-risk diagnostic step.

11. Serologic workup

WHAT YOU SEE

Workup toolbox of vials

WHAT IT ENCODES

After urinalysis, send targeted SEROLOGIES to pin the cause: C3/C4 complement, ANCA (anti-MPO/anti-PR3), anti-GBM antibody, ASO/anti-DNase B (post-strep), ANA and anti-dsDNA (lupus), cryoglobulins, hepatitis B/C, and HIV. The complement level is the pivotal first split (low vs normal).

BOARD TRAP

WRONG: 'Serologies alone can replace urinalysis as the initial glomerular workup.' Discriminator — serologies are step TWO, interpreted in light of the sediment; they refine the differential but do not substitute for the urinalysis that first establishes glomerular bleeding. Order both, with UA first.

12. Renal biopsy

WHAT YOU SEE

Biopsy needle / report on the clue board

WHAT IT ENCODES

Renal BIOPSY is the definitive step to classify the glomerular lesion (light microscopy, immunofluorescence, EM) when urinalysis plus serologies are inconclusive or when results would change management. It is generally pursued AFTER noninvasive workup — except suspected RPGN, where urgent biopsy is needed to start immunosuppression fast.

BOARD TRAP

WRONG: 'Renal biopsy should always be the first diagnostic test in suspected glomerulonephritis.' Discriminator — biopsy usually FOLLOWS urinalysis and serologies, since many cases (e.g., classic PSGN in a child) are diagnosed clinically. The major exception is suspected RPGN/crescentic disease, where you biopsy urgently rather than wait.